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B.Tech. Degree VIII Semester Examination April 2018

ME 1801 COMPRESSIBLE FLUID FLOW (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

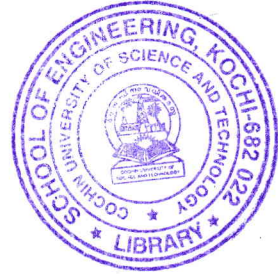
(Use of approved gas tables permitted)

PART A

(Answer **ALL** questions)

(8 × 5 = 40)

- I. (a) Briefly explain Mach cone and Mach angle.
- (b) Superheated steam is supplied to a supersonic nozzle at the design flow rate of 1.0 kg/s (per cm² of throat area). If the stagnation temperature of steam is 800 K, determine the stagnation pressure of the steam required for this flow rate.
Take $R = 0.477 \text{ kJ/kgK}$, $\gamma = 1.278$ for steam. Assume isentropic flow.
- (c) Compare Rankine Hugoniot and isentropic relations.
- (d) Briefly explain $\theta - \beta - M$ diagram.
- (e) Derive a relationship for velocity ratio for Fanno flow.
- (f) Briefly explain the slope of the Rayleigh line.
- (g) Briefly explain temperature recovery factor.
- (h) Explain the principle and working of Shadowgraph technique with the help of a neat sketch.



PART B

(4 × 15 = 60)

- II. (a) Derive expressions for determining pressure and temperature in troposphere. (5)
- (b) In a settling chamber air is at a pressure of 5 bar and a temperature of 500 K. Determine the following: (10)
 - (i) Stagnation enthalpy.
 - (ii) Stagnation velocity of sound.
 - (iii) Maximum velocity of fluid.
 - (iv) Critical temperature.
 - (v) Critical velocity of fluid and sound.

OR

- III. A supersonic wind tunnel settling chamber expands air or Freon – 21 through a nozzle from a pressure of 10 bar to 4 bar in the test section. Calculate the stagnation temperature to be maintained in the settling chamber to obtain a velocity of 500 m/s in the test section for : (15)
 - (i) Air, $C_p = 1.025 \text{ kJ/kgK}$, $C_v = 0.735 \text{ kJ/kgK}$.
 - (ii) Freon, $C_p = 0.785 \text{ kJ/kgK}$, $C_v = 0.675 \text{ kJ/kgK}$.

What is the test section Mach number in each case?

(P.T.O.)

- IV. The ratio of the exit to entry area in a subsonic diffuser is 4.0. The Mach number of a jet of air approaching the diffuser at a stagnation pressure of 1.013 bar and temperature of 290 K, is 2.2. There is a standing normal shock wave just outside the diffuser entry. The flow in the diffuser is isentropic. Determine at the exit of the diffuser. (15)
- Mach number.
 - Temperature.
 - Pressure.
 - What is the stagnation pressure loss between the initial and final states of the flow?
- OR**
- V. A jet of air at a Mach number of 2.5 is deflected inwards at the corner of a curved wall. The wave angle at the corner is 60° . Determine the deflection angle of the wall, pressure and temperature ratios and final Mach number. (15)
- VI. A gas ($\gamma = 1.3$, $R = 0.287$ kJ/kgK) at pressure 1.0 bar, temperature 400 K enters a 30 cm diameter duct at a Mach number of 2.0. A normal shock occurs at Mach number of 1.5 and the exit Mach number is 1.0. If the mean value of the friction factor is 0.003, determine : (15)
- Length of the duct upstream and downstream of the shock wave.
 - Mass flow rate of the gas.
 - Change of entropy upstream of the shock, across the shock, and downstream of the shock.
- OR**
- VII. The Mach number at the exit of a combustion chamber is 0.9. The ratio of stagnation temperatures at exit and entry is 3.74. If the pressure and temperature of the gas at exit are 2.5 bar and 1000°C respectively. Determine : (15)
- Mach number, pressure and temperature of the gas at the entry.
 - The heat supplied per kg of the gas and
 - The maximum heat that can be supplied.
- Take $\gamma = 1.3$ and $C_p = 1.218$ kJ/kgK.
- VIII. Explain the working of the following with help of neat sketches. (6)
- Shock tube. (6)
 - Constant temperature type hotwire anemometer. (9)
- OR**
- IX. Explain the working principles of the following with the help of neat sketches.
- Schlieren Technique. (7)
 - Subsonic wind tunnel. (8)
